

From Land to Liquidity: Compensation Design and Local Population Reallocation in Taiwan's Land-to-the-Tiller Reform

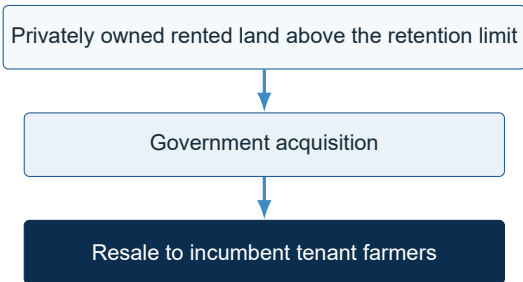
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Three stages of postwar land reform

Year	Name	Land or contractual relationship	Administrative action
1949	Rent Reduction	Private landlord–tenant contracts	Rent ceiling, tenure protection, and contract registration
1951	Sale of Public Land	Government-held cultivated land already under tenancy	Sale to incumbent cultivators with installment payment
1953	Land-to-the-Tiller	Privately owned rented land above legal retention limits	Government acquisition and resale to incumbent tenants

The 1953 Land-to-the-Tiller transaction



Compensation to former landlords

- ▶ 70% of land value in commodity-denominated land bonds
- ▶ 30% in shares of state-owned enterprises
- ▶ Bonds paid 4% annual interest and were amortized over ten years
- ▶ Outside Penghu, repayment was scheduled twice per year

Statutory text: bond denomination & repayment asset

本債券券面實物，分稻穀與甘薯二類：一、稻穀債券，定額十二億六千萬公斤，用以補償徵收田地目部份所包括之兩季田、單季田、看天田、輪作田（包括特殊田）地價。二、甘薯債券，定額四億四千萬公斤，用以補償徵收畑地目部份地價。以上二類債券，由臺灣省政府於民國四十二年度內按照債券面額十足發行。

本債券到期本息，依左列規定償付之：一、稻穀債券，償付稻穀。但用以補償輪作田（包括特殊田）地價所搭發之稻穀債券，按該年期當地稻穀時價折付現金。二、甘薯債券，按該年期之甘薯時價折付現金。前項現金本息，由臺灣土地銀行當地行處經付，其稻穀本息，得轉託糧食局指定之當地倉庫經付，其經付稻穀標準，依臺灣省田賦徵實實收規則之規定。

Article 4 links registered “田” categories to rice bonds and registered “畑” land to sweet-potato bonds. The kilogram totals report the bond denominations. Ordinary rice bonds delivered rice; the listed rotational/special-field exception and sweet-potato bonds were paid in cash at the relevant local current price.

Historical and administrative data

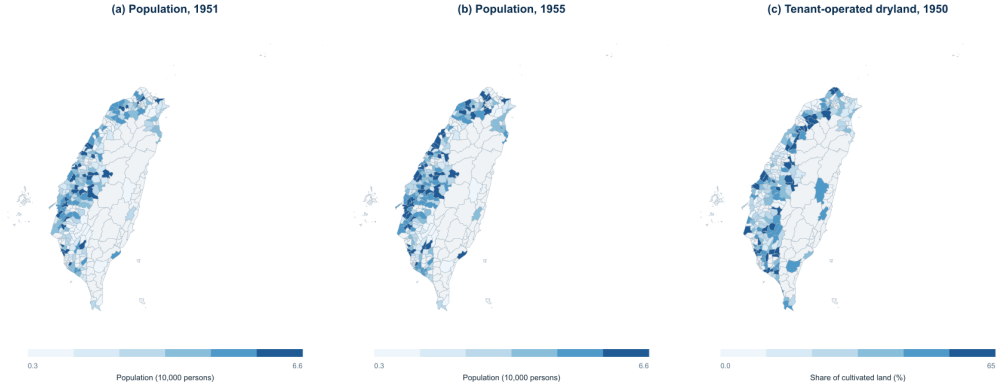
Years	Record	Original geography	Use in the analysis
1947–1954	County and municipal statistical yearbooks, statistical abstracts, local household-registration series, and local gazetteers	Township	Pre-reform population and population in 1954
1955–1958	《臺灣省戶籍統計要覽：民國三十五年至四十七年戶籍統計年報》	Township	Post-reform population
1915–1935	Colonial censuses and population statistics	Prefecture, district, street, and village	Prewar population
1950	《耕地所有權與經營狀況調查報告書》	Township	Pre-reform land ownership and categories
1950, 1953	Digitized research appendix by Chiang, Fan, and Hsu	Township	Cross-checks for parts of the 1950 land data and 1953 reform exposure
1953	《農復會工作報告》第五期	County and municipality	Actual paddy, dryland, and total area taken under the reform
1966	《中華民國 55 年臺閩地區戶口及住宅普查報告書》	Township	Supplementary time endpoints

Population panel and annual coverage

Year	1947	1948	1949	1950	1951	1952	1953	1954	1955	1956	1957	1958
Land-reporting units	148	149	154	205	237	237	227	237	237	237	237	237

- ▶ The 1950 land survey defines the 237 township-level units used in the main panel analysis.
- ▶ 2,542 unit–year cells are observed in the main panel.

Spatial variation in the analysis sample



Historical analysis units are displayed on 1982 township boundaries.

Empirical question and measurement

Question

Did local population paths change differentially in places more exposed to compensation that was repaid in cash?

Outcome

Registered local population in each township-level unit and year.

Baseline exposure

$$DryShare_i = \frac{\text{dryland area in 1950}}{\text{cultivated area in 1950}}$$

Alternative exposure

Dryland cultivated by tenants in 1950 and redicted redistributed dryland, both as a share of cultivated land, the latter constructed from 1950 owner-size bins.

Baseline continuous-treatment difference-in-differences

$$Population_{it} = \alpha_i + \lambda_t + \beta(DryShare_i \times Post_t) + \varepsilon_{it}$$

- ▶ $Post_t = 1$ for 1954 and later.
- ▶ Township fixed effects absorb time-invariant local differences.
- ▶ Year fixed effects absorb Taiwan-wide shocks.
- ▶ Standard errors are clustered by the 237 treatment units.

Baseline estimates and sample completeness

	All available (1)	95% coverage (2)	85% coverage (3)	Complete years (4)	Historical units (5)	Includes prewar (6)
$DryShare_i \times Post_t$	-1787.150** (830.791)	-1587.352** (670.739)	-1682.265** (698.707)	-1760.430** (743.458)	-1644.582* (899.257)	-7145.045*** (1862.641)
<i>p</i> -value	0.031	0.018	0.016	0.018	0.067	<0.001
Township FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,542	1,886	2,091	1,659	2468	4308
Treatment units	237	237	237	237	230	230
Sample years	1947–1958	1951–1958	1950–1958	7 complete years	1947-1958	Prewar and 1947-1958

Dependent variable: population count. Parentheses report standard errors clustered by treatment unit. * $p < 0.10$, ** $p < 0.05$,

*** $p < 0.01$.

Event-study estimates and pre-trend assessment



1953 is the omitted year; error bars are unit-clustered 95% confidence intervals. The 1951 estimate motivates the public-land specification reported next.

Step 1: observed inputs from the 1950 land survey

Private cultivated-land owners by holding-size bin

For township i , the survey reports n_{ib} owners in each of 12 ownership-area bins b :

$[0, 0.5)$, $[0.5, 1)$, $[1, 2)$, $[2, 3)$, $[3, 5)$, $[5, 7)$ ha
 $[7, 10)$, $[10, 20)$, $[20, 30)$, $[30, 50)$, $[50, 100)$, $[100, \infty)$ ha

$$N_i = \sum_{b=1}^{12} n_{ib}$$

is the number of owners represented in the grouped distribution.

Original source title: 《耕地所有權與經營狀況調查報告書》. The owner-size bins are measured in hectares; later calculations convert hectares and chia using $1 \text{ chia} = 0.969917 \text{ hectares}$.

Other township-level fields

- ▶ total cultivated area C_i
- ▶ paddy area and dryland area Dry_i
- ▶ cultivated area by operating status, including tenant-operated paddy and dryland

Step 2: fit a township owner-size distribution

Recommended distribution

$$F_i(a) = \Pr(A_i \leq a) = \Phi\left(\frac{\ln a - \mu_i}{\sigma_i}\right)$$

County-bin shares $\pi_{c(i)b}$ supply limited shrinkage for small townships:

$$\tilde{n}_{ib} = n_{ib} + \kappa \pi_{c(i)b}, \quad \kappa = 25.$$

Parameters are estimated from the grouped likelihood:

$$(\hat{\mu}_i, \hat{\sigma}_i) = \arg \max_{\mu, \sigma} \sum_b \tilde{n}_{ib} \ln[F_i(u_b) - F_i(l_b)].$$

What the shrinkage does

- ▶ Adds 25 pseudo-owners distributed according to the 1950 county bin shares.
- ▶ Stabilizes the bin proportions of townships with few observed owners.
- ▶ Uses the 1950 land survey; N_i and fitted excess area determine each township estimate.

Model comparison

Five candidates are compared: lognormal, log-logistic, gamma, generalized gamma, and lognormal with a Pareto tail. The selected specification is lognormal with $\kappa = 25$.

The township likelihood uses the 1950 survey inputs. The 1953 county totals enter model comparison and validation.

Step 3: construct treatment intensity

Expected land above a uniform threshold

Let $h = 3$ chia (2.909751 hectares). For one owner,

$$e_i = \mathbb{E}[(A_i - h)_+] = \int_h^\infty [1 - F_i(a)] da.$$

For a lognormal distribution,

$$e_i = \exp\left(\hat{\mu}_i + \frac{\hat{\sigma}_i^2}{2}\right) \Phi\left(\frac{\hat{\mu}_i + \hat{\sigma}_i^2 - \ln h}{\hat{\sigma}_i}\right) - h\Phi\left(\frac{\hat{\mu}_i - \ln h}{\hat{\sigma}_i}\right).$$

Predicted township redistribution is

$$\hat{R}_i = N_i e_i,$$

after converting the result back from hectares to chia.

Dryland component and normalization

$$q_i = \frac{Dry_i}{C_i}, \quad \hat{R}_i^D = \hat{R}_i q_i,$$

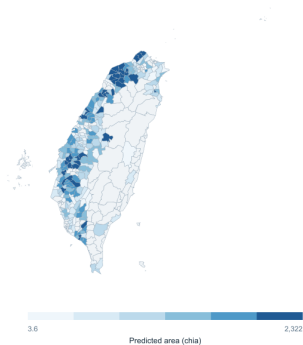
$$D_i = \frac{\hat{R}_i^D}{C_i}.$$

An alternative replaces q_i with the dryland share of tenant-operated cultivated land.

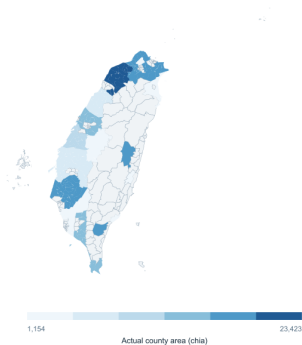
D_i is the regression treatment: predicted redistributed dryland divided by township cultivated land.

Predicted and actual redistribution across space

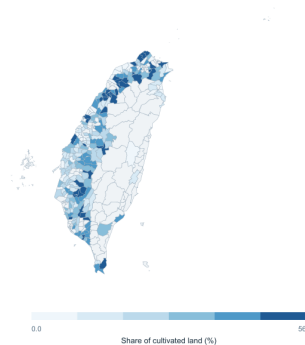
(a) Predicted redistribution



(b) Actual redistribution, 1953

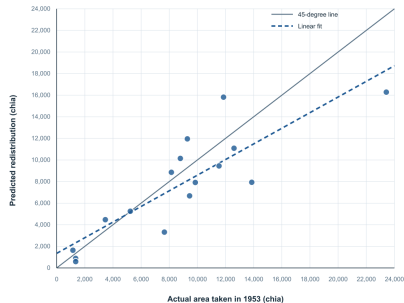


(c) Predicted dryland component



Validation against 1953 county totals

Construct	Predicted area	Actual area	Predicted / actual	Level corr.	Log corr.
Total redistributed land	122,313	139,003	0.880	0.852	0.923
Dryland: township land mix	37,056	21,385	1.733	0.722	0.856
Dryland: tenant land mix	30,204	21,385	1.412	0.800	0.886



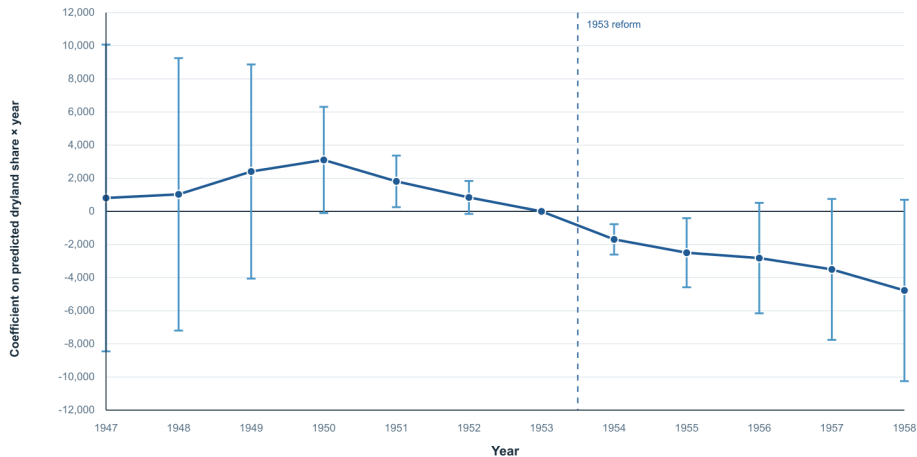
Population estimates using predicted redistribution

	(1) Total share	(2) Dry share: township mix	(3) Dry share: tenant mix
$\hat{R}_i / C_i \times Post_t$	2091.892 (1643.544)		
$\hat{R}_i^D / C_i \times Post_t$		-4343.526* (2418.013)	-7875.207*** (2593.994)
Township fixed effects	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
Observations / units	2,542 / 237	2,542 / 237	2,542 / 237

Total-share coefficient: $p = 0.203$.

Dry-share coefficients: $p = 0.072$ and $p = 0.002$.

Event-study estimates using predicted redistributed dryland



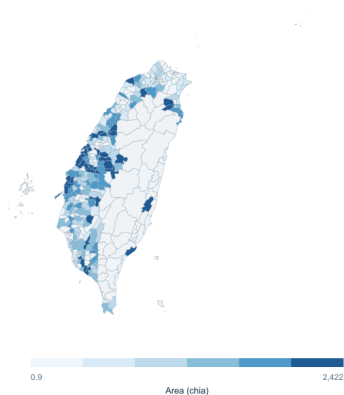
Total reform intensity and dryland horse-race

	(1) Total intensity	(2) Total intensity and dryland composition
$T_i \times Post_t$	2091.892 (1643.544)	1931.320 (1575.405)
$Q_i \times Post_t$		-1686.137** (808.783)
Unit / year fixed effects	Yes	Yes
Observations / units	2,542 / 237	2,542 / 237

$T_i = \hat{R}_i / C_i$ is total reform intensity. $Q_i = \hat{R}_i^D / \hat{R}_i$ is the dryland share of predicted redistributed land. In column (2), $p = 0.220$ for total intensity and $p = 0.037$ for dryland composition.

Potential confound: the 1951 sale of public land

Leased public cultivated land



Institutional distinction

- ▶ The 1951 program covered government-held cultivated land.
- ▶ Incumbent cultivators purchased the land in installments.
- ▶ Its timing predates the 1953 acquisition of privately owned rented land.

Empirical proxy

Leased public cultivated land in the 1950 survey, measured in chia and transformed using a standardized inverse hyperbolic sine.

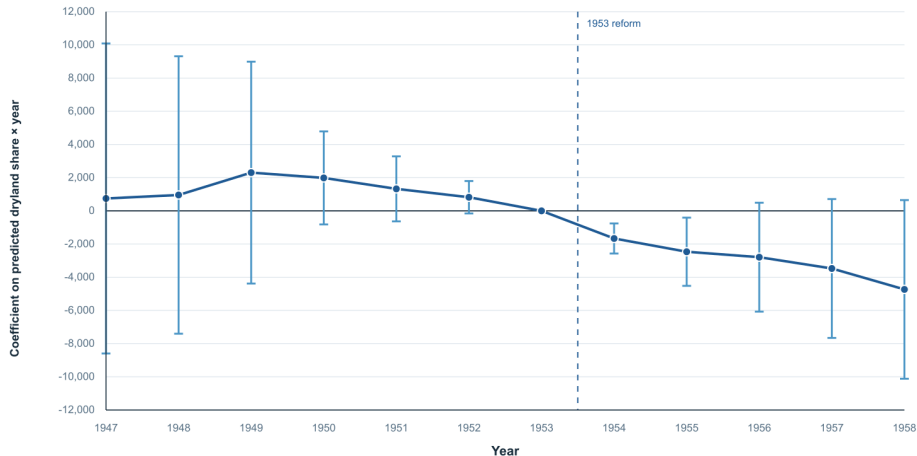
Joint specification for the two reform exposures

Exposure interaction	Coefficient	Clustered SE	p-value	Scale
$PredDry_i \times 1[t \geq 1954]$	-3999.975*	2427.082	0.099	10 pp: -400.0 persons
$PublicLease_i \times 1[t \geq 1952]$	551.698***	172.889	0.001	1 SD: 551.7 persons
Observations / treatment units		2,542 / 237		
Unit / year fixed effects		Yes		

$PredDry_i$ is predicted redistributed dryland divided by cultivated land.

$PublicLease_i$ is the standardized inverse-hyperbolic-sine measure.

Event study with public-land exposure controlled



Pre-reform joint test $\chi^2(6) = 5.838$,
 $p = 0.442$

Post-reform joint test $\chi^2(5) = 22.966$,
 $p < 0.001$

Conclusion and contributions

1. A 10-percentage-point increase in the 1950 dryland share reduces local population by 178.7 residents after 1954 ($p = 0.031$). After controlling separately for exposure to the 1951 Sale of Public Land, the pre-reform event-study joint test yields $p = 0.442$.
2. The estimates provide quantitative causal evidence that the reform facilitated rural population outflows and economic modernization by unlocking liquidity previously tied up in land.
3. This study constructs two new archival datasets: an annual household-registration panel and a township-level cross-section of cultivated landholdings.
4. The findings place land finance and structural modernization at the center of the relationship between land reform and Taiwan's postwar economic takeoff, shifting attention beyond the conventional productivity-centered account.